CENTRAL POWER RESEARCH INSTITUTE

CPRI

TEST REPORT

Test Report Number	:	DMDLDL16-17S0117/N Date: 07.10.2016
Name & Address of the Customer	:	M/s Savita Oil Polymers Ltd, 66/67, Nariman Bhavan , 6th floor Nariman Point, Mumbai - 400 021.
	:	Customer's Reference and date Letter dated 09.09.2016 & email dtd 04.10.2016
Name & Address of the Manufacturer	:	M/s Savita Oil Polymers Ltd, Mumbai - 400 021.
Particulars of Sample tested	:	Biotransol - HF
Condition of Sample on Receipt	:	New.
Type	:	Not applicable
Description of sample	:	Biotransol -HF
Serial Number	:	Refer "Sheet 2 of 4 "
Number of Samples Tested	:	One only.
Samples received on	:	14.09.2016.
Date(s) of Test(s)	:	27.09.2016 to 06.10.2016
CPRI Sample Code No.	:	DMDLDL16-17S0117/N
Particulars of tests conducted	:	Refer "Sheet 2 to 3 of 4"
Tests in accordance with standard/Specification	:	IEC 62770 -2013
Sampling Plan	:	Nil
Customer's requirement	:	As indicated in "Sheet 2 to 3 of 4".
Deviations if any	:	Nil
Name of the witnessing persons	:	
Customer's representatives	:	None
Other than customer's representatives	:	None
Test subcontracted with address of the laboratory	:	Nil
Documents constituting this report (in words)	:	
Number of Sheets	:	Four only
Number of Oscillograms	:	Nil
Number of Graphs	:	Nil
Number of photos	:	Nil
Number of test circuit diagrams	:	Nil
Number of Drawings	:	Nil

(Ann Pamla Cruze)
Test Engineer

(V.V. Pattanshetti) 13/10/16
Head of Division

DIELECTRIC MATERIALS DIVISION
P.B. NO. 8066, SADASHIVANAGAR P.O.
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SAMPLE IDENTIFICATION

Sample of natural ester in 2.5 liter brown bottle duly sealed with tape and labeled as BIOTRANSOIL -HF

Sl. No.	Characteristics	IEC 62770 - Specification	Results	Remarks/Test Method
	PHYSICAL			
1.	Appearance	Clear, free from sediment and suspended matter	Clear, free from sediment and suspended matter	IEC: 62770
2.	Viscosity @ 40 °C, mm^2/s	50 (max)	33.45	ISO: 3104-1994
3.	Viscosity @ 100 °C, mm^2/s	15 (max)	7.730	ISO: 3104 -1994
4.	Pour point, °C	-10 (min)	-15	ISO: 3016 -1994 Automatic instrument used
5.	Water content, mg/kg	200 (max)	46	IEC: 60814 -1997
6.	Density @ 20 °C, g/ml	1.0 (max)	0.9210	ISO: 3675 -1976
	ELECTRICAL			
7.	a) Breakdown voltage, kV b) Mean of the Breakdown voltages, kV c) Breakdown voltage obtained during each test, kV	35 (min)	84 84.0 91.2,84.4,74.5, 96.1,78.3,79.4	IEC: 60156-1995 a) Type of electrodes: Brass spherical b) Frequency of the test voltage : 49.80 Hz c)Temperature of the liquid,: 24.2°C d) Use of stirrer (if any): yes

(Ann Pamla Cruze)
Test Engineer

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Sl. No.	Characteristics	IEC 62770 – Specification	Results	Remarks/Test Method
8.	Dielectric Dissipation Factor at 90°C	0.05 (max)	0.031	IEC: 60247 -2004 (i) Electrical stress: 250 V AC /mm (ii) Frequency of applied voltage:49.79Hz
	CHEMICAL			
9.	Total Acidity, mg KOH/g	0.06(max)	0.031	IEC: 62021-1:2003 Potentiometric method
10.	Corrosive Sulphur,(PCS)	Non-Corrosive	Non-Corrosive	IEC: 62535 - 2008
11.	Corrosive Sulphur	Non-Corrosive	Non-Corrosive	ASTM D 1275 B
	PERFORMAMCE			
12.	Oxidation Stability			IEC:61125 -1992 Method (C): 48 Hrs
	(a)Kinematic Viscosity @ 40 °C, mm^2/s	30% increase over initial value	40.46	
	(b)Total Acidity, mg KOH/g	0.6(max)	0.080	
	(c)Dissipation factor at 90°C	0.5(max)	0.16	IEC: 60247 -2004 (i) Electrical stress: 250 V AC /mm (ii) Frequency of applied Voltage: 49.75Hz
	HEALTH SAFETY & ENVIRONMENT			
13.	Flash Point, °C	250(min)	312	ISO:2592 - 2009
14.	Fire Point , °C	300(min)	360	ISO:2592 - 2009

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